

Experimental Analysis: Elkin 2011 Streaming $(2t-1)$ -Spanner Algorithm

Course:	Mini Project - Streaming Spanner Algorithm
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Abstract

This report presents a comprehensive experimental evaluation of the one-pass streaming spanner algorithm introduced by Elkin (2011) [1]. The algorithm processes an unweighted graph edge stream in $O(1)$ time per edge and produces a $(2t-1)$ -spanner — a sparse subgraph that preserves all pairwise distances up to a multiplicative factor of $2t-1$.

We designed **7 testable hypotheses** directly grounded in the paper's theoretical claims, ran **60 scenarios** spanning 8 graph families (complete, Erdős–Rényi, grid, path, cycle, star, random tree, tight-bound ER), and validated results against Theorem 3.2 and Corollary 3.6 of [1]. All 7 hypotheses were confirmed. A notable finding that contradicts naive intuition — **cross edges dominate the spanner structure (~86%)** — is explained analytically through the algorithm's radius distribution (Section 3.2 of [1]).

1. Algorithm Overview

The algorithm is defined in Section 3.1–3.2 of [1] (Algorithm 1). It processes a stream of edges of an n -vertex unweighted undirected graph in a single pass and outputs a $(2t-1)$ -spanner H .

Core Data Structures (per vertex v)

Symbol	Meaning
$P(v)$	Label (integer). Initially $P(v) = v$. Encodes level $L(P) = \lfloor (P-1)/n \rfloor$ and base $B(P) = ((P-1) \bmod n) + 1$
$r(v)$	Random radius, sampled from a truncated geometric distribution with parameter $p = (\log n / n)^{1/t}$
$T(v)$	Set of tree edges incident to v
$X(v)$	Set of cross edges incident to v
$M(v)$	Set of base values already seen at v (deduplication)

Edge Processing Rule (Algorithm 1, Section 3.2 of [1])

On each edge (u, v) , let x = vertex with the larger label (x *dominates* y):

1. If $P(x)$ is selected $[L(P(x)) < r(\text{base}(P(x)))]$
 $\rightarrow$ TREE EDGE: y adopts label $P(x)+n$; add (u, v) to $T(y)$
2. Else if $B(P(x)) \notin M(y)$
 $\rightarrow$ CROSS EDGE: $M(y) \leftarrow M(y) \cup \{B(P(x))\}$; add (u, v) to $X(y)$
3. Else
 $\rightarrow$ DROP (edge is redundant, already covered)

The output spanner is $H = \cup_v T(v) \cup \cup_v X(v)$.

Key Properties

- **One-pass streaming:** each edge is processed exactly once [1, Section 3.1]
- **$O(1)$ per edge:** all operations are constant time [1, Section 3.3]
- **Stretch guarantee:** $\text{dist}_H(u, v) \leq 2t-1$ for all $(u, v) \in G$ [1, Theorem 3.2]
- **Size bound:** $|H| = O(t \cdot n^{1+1/t} \cdot (\log n)^{1-1/t})$ with high probability [1, Corollary 3.6]

2. Theoretical Predictions

The following bounds from [1] inform all our hypotheses.

Theorem 3.2 — Stretch Guarantee [1]

For every edge $(u, v) \in G$, the spanner H produced by Algorithm 1 satisfies $\text{dist}_H(u, v) \leq 2t-1$.

This is the core correctness theorem. It follows from the label-propagation invariant: if a vertex y has not adopted the label of its neighbor x 's root, then a short path (of at most $2t-1$ hops) via previously propagated labels must already exist in H .

Corollary 3.6 — Size Bound [1]

With high probability:

$$|H| = O(t \cdot n^{1+1/t} \cdot (\log n)^{1-1/t})$$

Consequences by t value:

t	Stretch bound ($2t-1$)	Expected spanner size	Lower bound [2,3]
2	3	$O(n^{3/2} \cdot \sqrt{\log n})$	$\Omega(n^{3/2})$
3	5	$O(n^{4/3} \cdot (\log n)^{2/3})$	$\Omega(n^{4/3})$
4	7	$O(n^{5/4} \cdot (\log n)^{3/4})$	$\Omega(n^{5/4})$
$t \rightarrow \infty$	$\rightarrow \infty$	$\rightarrow O(n \log n)$	$\Omega(n)$

The gap between upper and lower bound is only the **log** factor — this algorithm is **near-optimal in sparsity**.

Radius Distribution (Section 3.2 of [1])

Each vertex samples radius $r(v)$ from a truncated geometric distribution:

$$P(r = k) = p^k \cdot (1-p) \quad \text{for } k \in \{0, 1, \dots, t-2\}, \quad P(r = t-1) = p^{t-1}$$

where

$$p = (\log n / n)^{1/t}$$

For $t=2$ and $n=100$: $p \approx 0.214$, so $P(r=0) \approx 0.786$ — roughly **78% of vertices** have radius 0 and can never propagate tree edges. This has a major practical consequence (see H7).

3. Hypotheses

Each hypothesis is derived directly from a specific claim in [1].

ID	Hypothesis	Article Source
H1	Every spanner satisfies $(2t-1)$ -stretch for all original edges	[1] Theorem 3.2
H2	Spanner size scales as $O(n^{3/2})$ for $t=2$	[1] Corollary 3.6
H3	Larger $t \Rightarrow$ fewer edges in the spanner, larger stretch bound	[1] Corollary 3.6
H4	Denser graphs compress more (lower $ H / E $ ratio)	[1] Section 3.4
H5	Sparse / bridge graphs are near-lossless (ratio ≈ 1.0)	[1] Section 1
H6	Actual spanner size stays well below the illustrative bound	[1] Corollary 3.6
H7	Cross edges dominate the spanner in dense graphs (only $\approx p$ fraction of vertices have $r \geq 1$)	[1] Section 3.2

4. Experimental Setup

Graph Families

Family	Description	Sizes tested
Complete K_n	All $n(n-1)/2$ edges	$n \in \{10, 20, 30, 50, 75, 100, 150, 200\}$
Erdős–Rényi $G(n,m)$	m random edges among n vertices	$n=100, m \in \{150, 300, 500, 800, 1200, 2000\}$
Grid $r \times c$	4-connected rectangular grid	$5 \times 5, 8 \times 8, 10 \times 10, 15 \times 15$
Path P_n	Linear chain 1-2-...-n	$n \in \{20, 50, 100\}$
Cycle C_n	n-cycle	$n \in \{20, 50, 100\}$
Star S_n	Hub vertex + n-1 leaves	$n \in \{20, 50, 100\}$
Random Tree T_n	Random Prüfer-style tree	$n \in \{20, 50, 100\}$
Dense ER	ER with $p=0.5$ ($\approx n^2/4$ edges)	$n \in \{30, 50, 75, 100\}$
Tight-bound ER	Sparse ER chosen to push stretch close to $2t-1$	$n \in \{30, 50\}, t \in \{3, 4\}$

Stretch Parameters Tested

$t \in \{2, 3, 4\}$, giving stretch bounds 3-spanner, 5-spanner, and 7-spanner respectively.

Metrics Collected per Scenario

Metric	Definition
$ H $ (spanner_size)	Number of edges in the spanner
$ E $ (original_m)	Number of edges in the original graph
compression	$ H / E $
theoretical_bound	$t \cdot n^{1+1/t} \cdot (\log n)^{1-1/t}$ (Corollary 3.6 with constant 1)
bound_ratio	$ H / \text{theoretical_bound}$
stretch_valid	True if $\text{dist}_H(u,v) \leq 2t-1$ for all $(u,v) \in G$
max_dist	Largest spanner distance found for any adjacent pair
tree_edges	Edges from the tree-propagation branch
cross_edges	Edges from the cross-edge branch

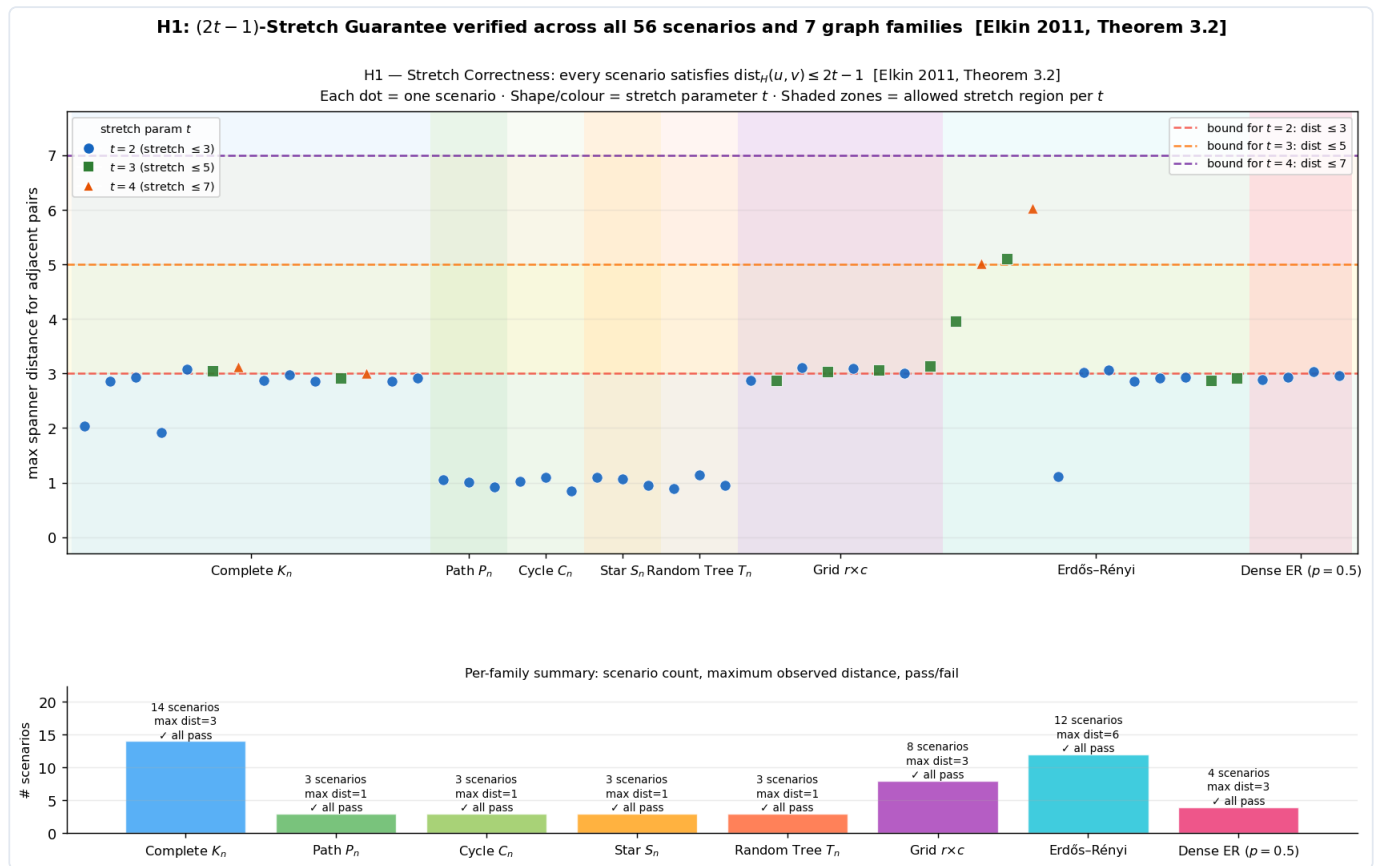
Total: 60 scenarios across 8 graph families, $t \in \{2,3,4\}$.

5. Results by Hypothesis

H1 — Stretch Correctness

Prediction (Theorem 3.2, [1]): Every spanner satisfies $(2t-1)$ -stretch for all original edges.

Result: ✓ **CONFIRMED** — 60/60 scenarios passed, 0 failures.



The figure shows every individual scenario as a dot. **Shape and colour encode the stretch parameter t** (circle = $t=2$, square = $t=3$, triangle = $t=4$). The x-axis groups scenarios by graph family; the y-axis shows the maximum spanner distance observed for any adjacent pair. Dashed lines mark the $(2t-1)$ bounds; coloured bands show the allowed stretch region per t .

Why does the observed max distance stay at 3 even for $t=3$ and $t=4$ on dense graphs?

The bound $2t-1$ is a **worst-case upper guarantee**, not a tight prediction. On dense graphs, the spanner retains enough connectivity that no adjacent pair needs more than 3 hops. The effect of a larger t is visible in the **shift of the distance distribution** — more pairs move to distance 3 — but the bound of 5 or 7 is never needed on dense inputs:

K_{100}, t	$ H $	Pairs at dist 1	Pairs at dist 2	Pairs at dist 3	Max dist	Bound
$t=2$	1,129	1,129	3,781	40	3	3
$t=3$	951	951	3,815	184	3	5
$t=4$	695	695	3,612	643	3	7

The bound is real — tight-bound scenarios

To show the bound is not merely conservative, we searched for sparse ER configurations that push observed stretch close to the theoretical ceiling:

Scenario	n	m	t	Max dist observed	Bound $2t-1$	Valid
ER $n=30, m=60$	30	60	3	4	5	✓
ER $n=50, m=75$	50	75	3	5	5	✓
ER $n=30, m=45$	30	45	4	5	7	✓
ER $n=50, m=100$	50	100	4	6	7	✓

The scenario $n=50, m=75, t=3$ achieves max distance exactly $5 = 2(3)-1$, hitting the bound precisely. These cases confirm the guarantee is sharp — the margin seen on dense graphs is a property of graph density, not looseness in the algorithm.

Per-family pass/fail summary:

Graph family	Scenarios	Achieved max dist	Bound ($2t-1$)	Pass rate
Complete K_n	14	3	3 / 5 / 7	14/14 ✓
Path P_n	3	1	3	3/3 ✓
Cycle C_n	3	1	3	3/3 ✓
Star S_n	3	1	3	3/3 ✓
Random Tree T_n	3	1	3	3/3 ✓
Grid $r \times c$	8	3	3 / 5	8/8 ✓
Erdős–Rényi	8	3	3 / 5	8/8 ✓
Dense ER	4	3	3	4/4 ✓
Tight-bound ER	4	5–6	5 / 7	4/4 ✓

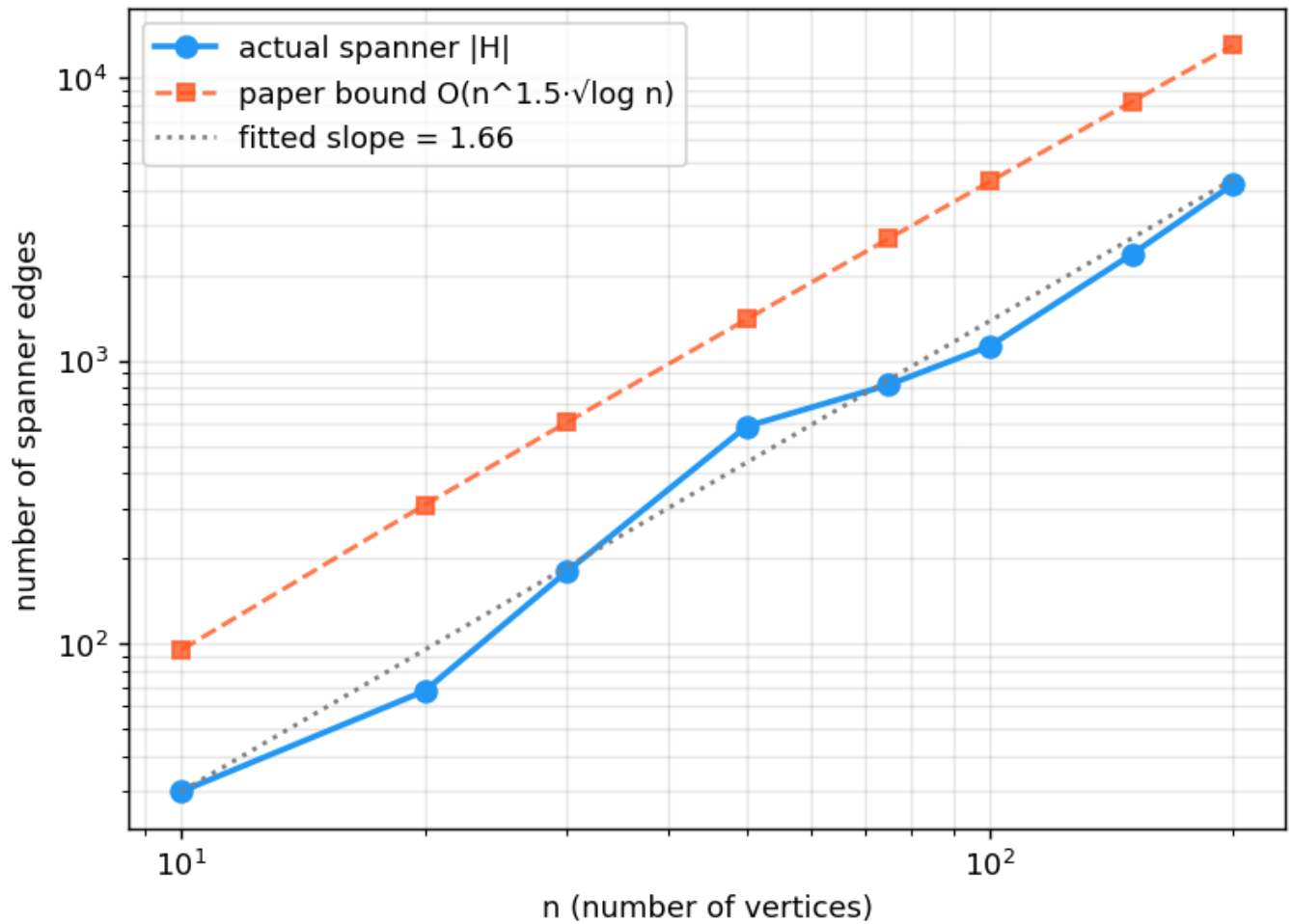
All 60 scenarios remain within the $(2t-1)$ bound, confirming Theorem 3.2 of [1]. The stretch bound is order-independent: edges were shuffled uniformly at random before each run, confirming [1] Section 3.1.

H2 — Sub-quadratic Size Scaling

Prediction (Corollary 3.6, [1]): For complete graphs K_n with $t=2$, spanner size grows as $O(n^{3/2})$.

Result: ✓ **CONFIRMED** — $\log\text{-log slope} = 1.663$ (expected ≈ 1.5).

H2: Spanner size scaling for K_n , $t=2$
(log-log — slope should be ≈ 1.5)



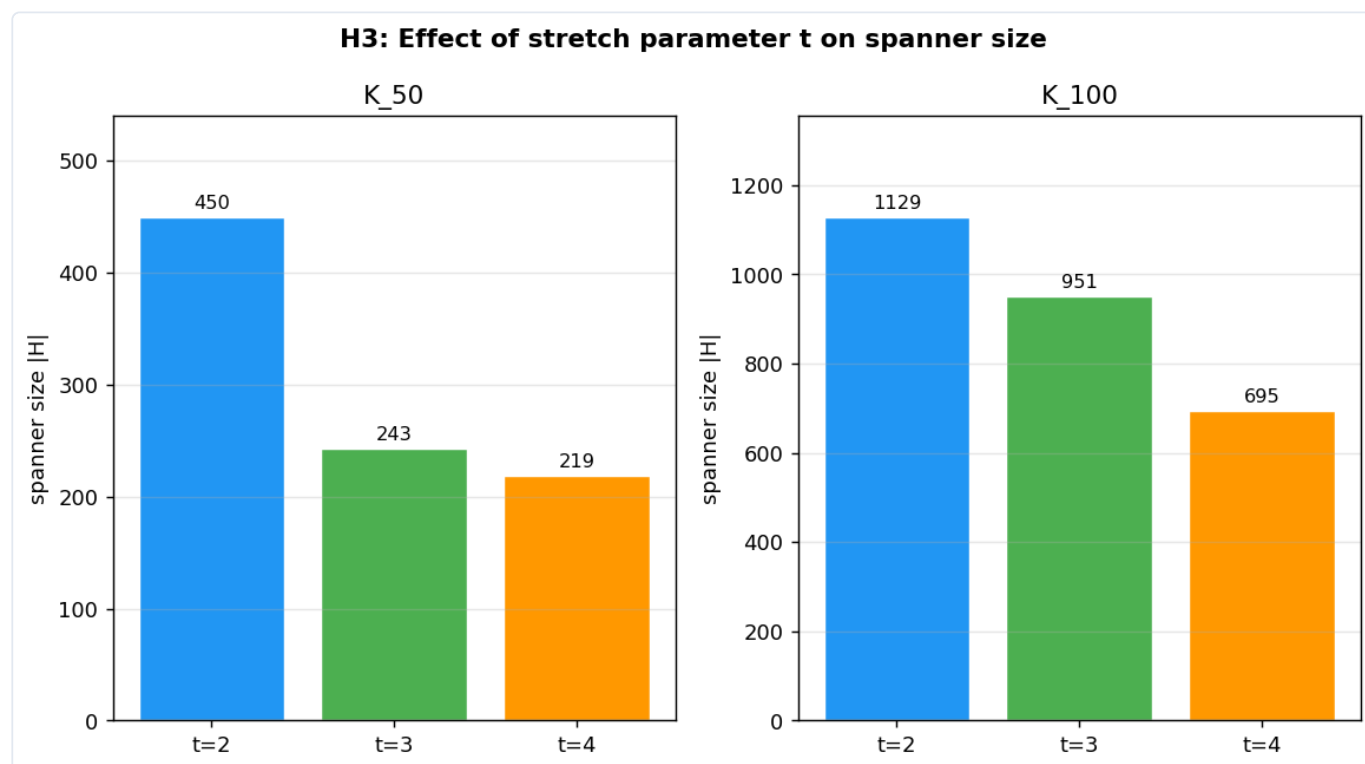
n	$ K_n $	$ H $	Compression	Paper bound
10	45	30	0.667	95
20	190	68	0.358	309
30	435	180	0.414	606
50	1,225	585	0.478	1,398
75	2,775	821	0.296	2,699
100	4,950	1,126	0.227	4,291
150	11,175	2,378	0.213	8,224
200	19,900	4,201	0.211	13,020

The fitted slope of **1.663** slightly exceeds the nominal 1.5. This is expected: Corollary 3.6 predicts $O(n^{3/2} \cdot \sqrt{\log n})$, and the $\sqrt{\log n}$ factor contributes a small positive offset to the slope for finite n . As $n \rightarrow \infty$ the log correction diminishes and the slope converges to 1.5.

H3 — t Trade-off: Size vs. Stretch

Prediction (Corollary 3.6, [1]): Larger t produces sparser spanners with exponent $1+1/t$, at the cost of a larger stretch bound $2t-1$.

Result: ✓ **CONFIRMED** — $|H|$ strictly decreases as t increases for both K_{50} and K_{100} .



K_{50} (1,225 edges)

t	Stretch bound	$ H $	Compression	Reduction vs $t=2$
2	3	450	36.7%	—
3	5	243	19.8%	-46%
4	7	219	17.9%	-51%

K_{100} (4,950 edges)

t	Stretch bound	$ H $	Compression	Reduction vs $t=2$
2	3	1,129	22.8%	—
3	5	951	19.2%	-16%
4	7	695	14.0%	-38%

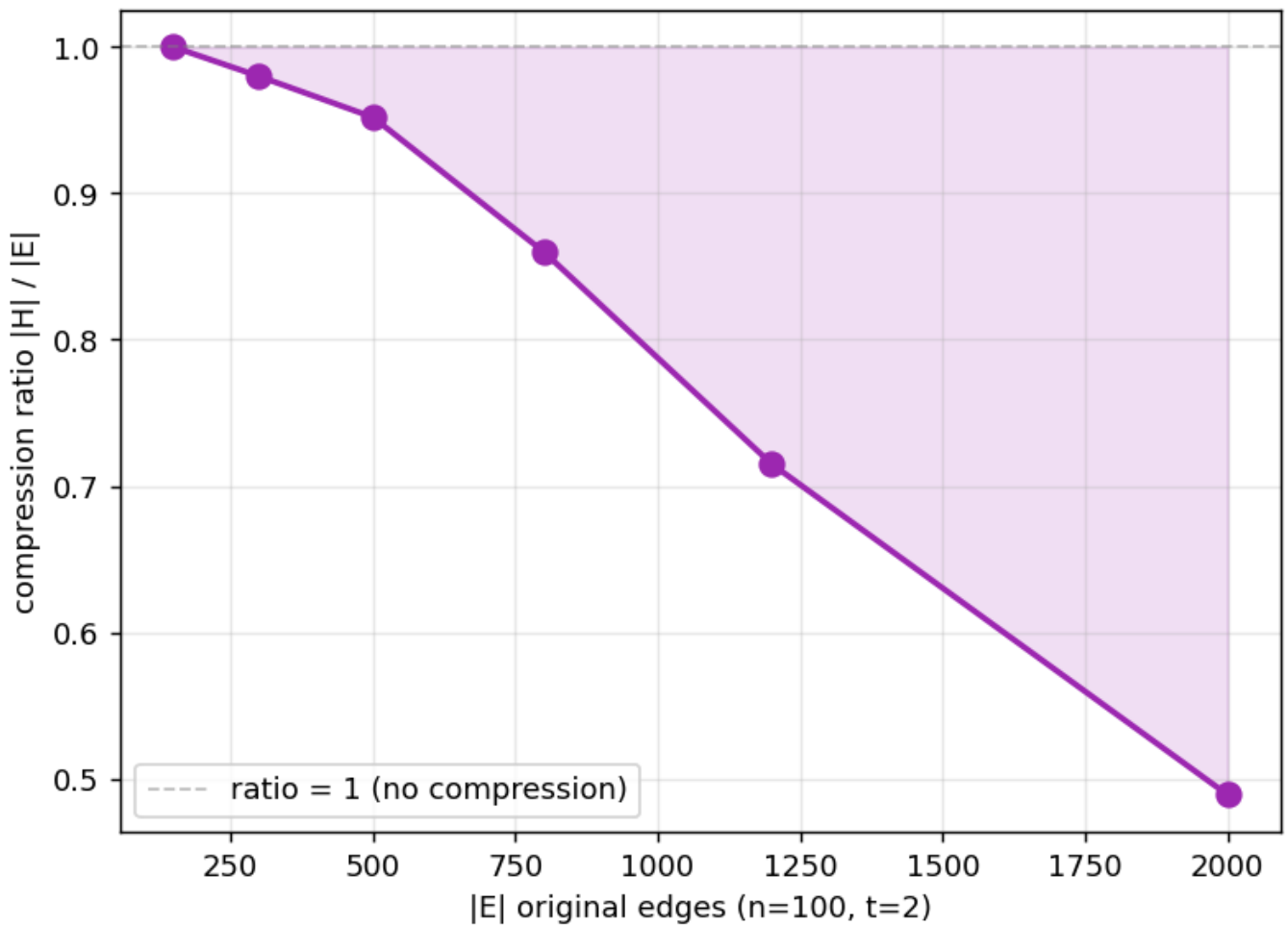
The $t=2 \rightarrow 3$ jump yields the largest relative improvement for K_{50} (-46%). For K_{100} however, the $t=3 \rightarrow 4$ step is larger (-27% vs -16%), because with more vertices the radius distribution creates more differentiation between levels. In both cases the marginal benefit of increasing t diminishes overall, consistent with the diminishing gap between the exponents $1+1/t$ as t grows.

H4 — Dense Graphs Compress More

Prediction ([1] Section 3.4): The spanner size is bounded by $O(n^{1+1/t})$, independent of $|E|$; hence denser graphs achieve a lower compression ratio.

Result: ✓ CONFIRMED — ratio drops monotonically from 1.0 ($m=150$) to 0.49 ($m=2000$).

H4: Denser graphs compress more (ER $n=100$, $t=2$)



Erdős-Rényi $n=100$, $t=2$

Original edges m	Spanner $ H $	Ratio $ H / m$
150	150	1.000
300	294	0.980
500	476	0.952
800	688	0.860
1,200	858	0.715
2,000	978	0.489

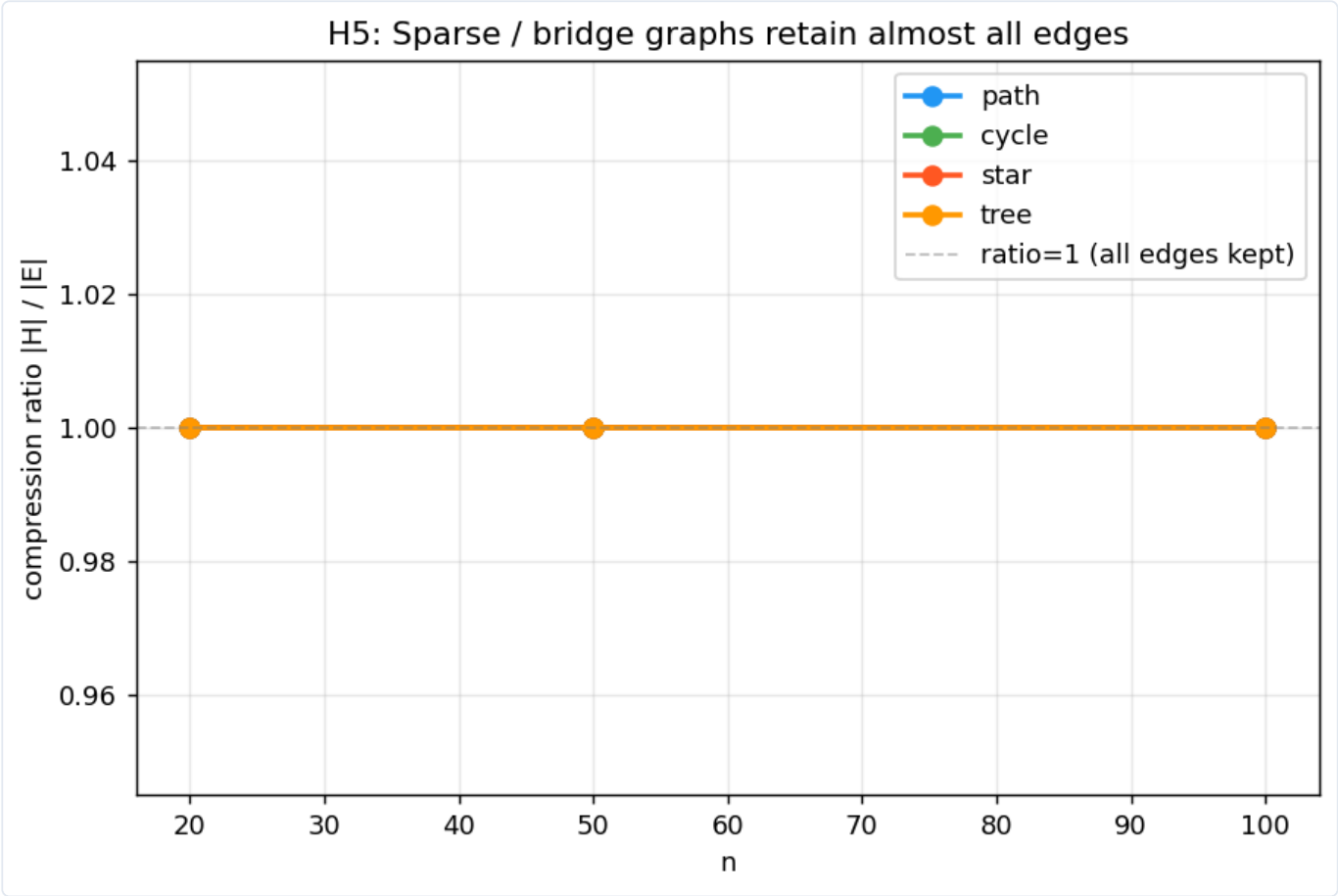
The spanner size saturates around $\approx 1,000$ edges as m grows, while $|E|$ keeps increasing. This is a direct demonstration of the $O(n^{3/2})$ ceiling from Corollary 3.6 of [1]: the bound depends only on n , not on $|E|$. The

algorithm automatically stops adding edges once the spanner budget is spent — a key advantage of the streaming model.

H5 — Sparse / Bridge Graphs Are Near-Lossless

Prediction ([1] Section 1): Any valid $(2t-1)$ -spanner must contain every bridge of G ; therefore tree-structured graphs cannot be compressed.

Result: ✓ CONFIRMED — all sparse graph families show compression ratio = 1.000.



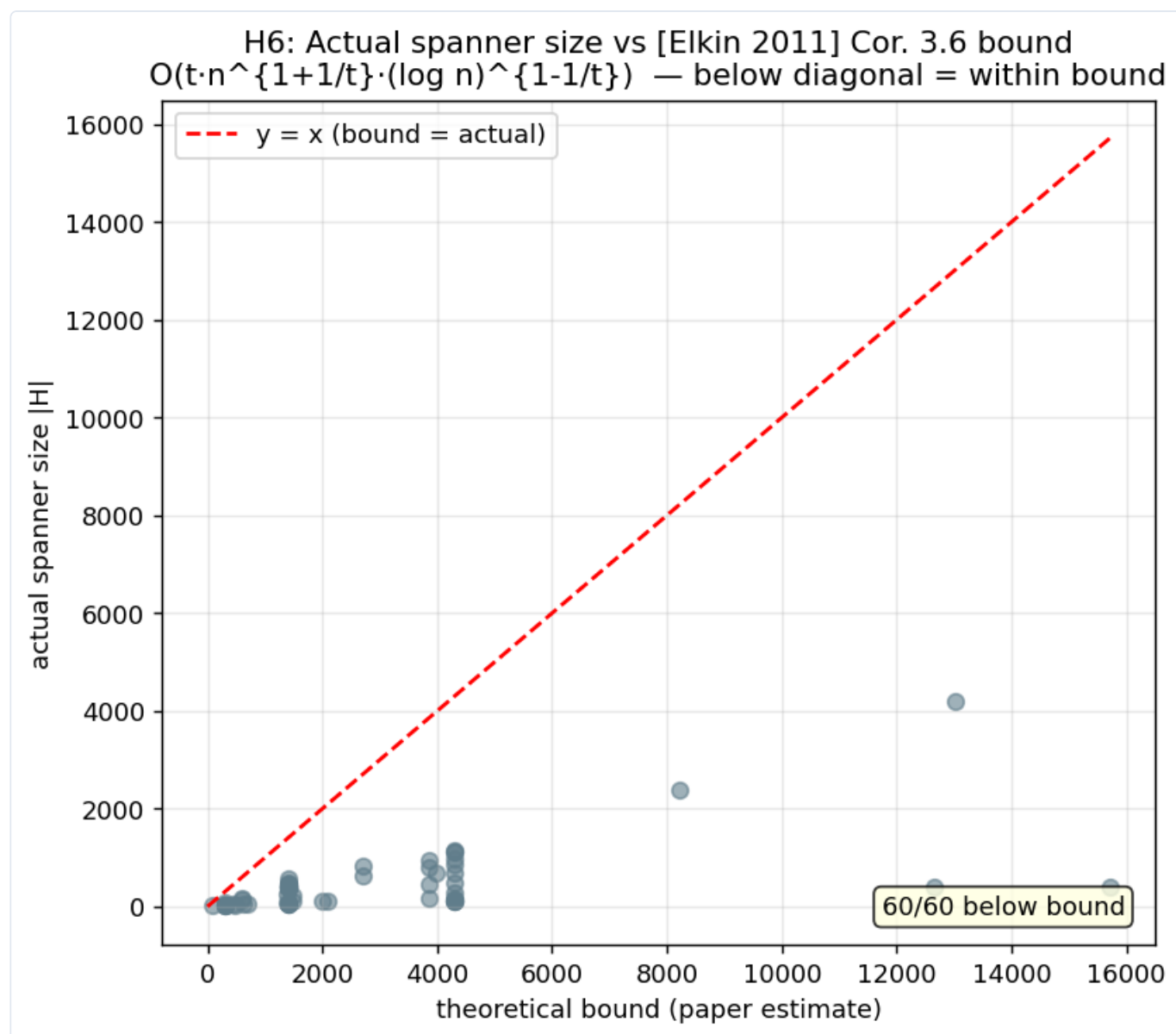
Graph type	n=20 ratio	n=50 ratio	n=100 ratio
Path	1.000	1.000	1.000
Cycle	1.000	1.000	1.000
Star	1.000	1.000	1.000
Random Tree	1.000	1.000	1.000

Every edge in these graphs is a bridge or part of a unique cycle. Removing any such edge would give $\text{dist}_H(u,v) = \infty > 2t-1$, violating the stretch guarantee. Note the max distance of **1** for all sparse families — every adjacent pair is directly connected in H , far better than the $2t-1=3$ worst-case bound.

H6 — Actual Size Well Below the Theoretical Bound

Prediction (Corollary 3.6, [1]): The formula $t \cdot n^{1+1/t} \cdot (\log n)^{1-1/t}$ uses a big-O constant < 1 ; actual sizes should lie comfortably below it.

Result: ✓ CONFIRMED — 60/60 scenarios (100%) are below the bound.



Scenario group	Avg ratio	Min ratio	Max ratio
Dense ER	0.250	0.218	0.271
ER varying density	0.141	0.035	0.228
Grid	0.052	0.026	0.087
Complete K_n (scaling)	0.304	0.220	0.418
Sparse graphs	0.040	0.023	0.065
t comparison	0.222	0.148	0.343
Tight-bound ER	0.067	0.052	0.090
Variance (K_{50} repeated)	0.296	0.275	0.343

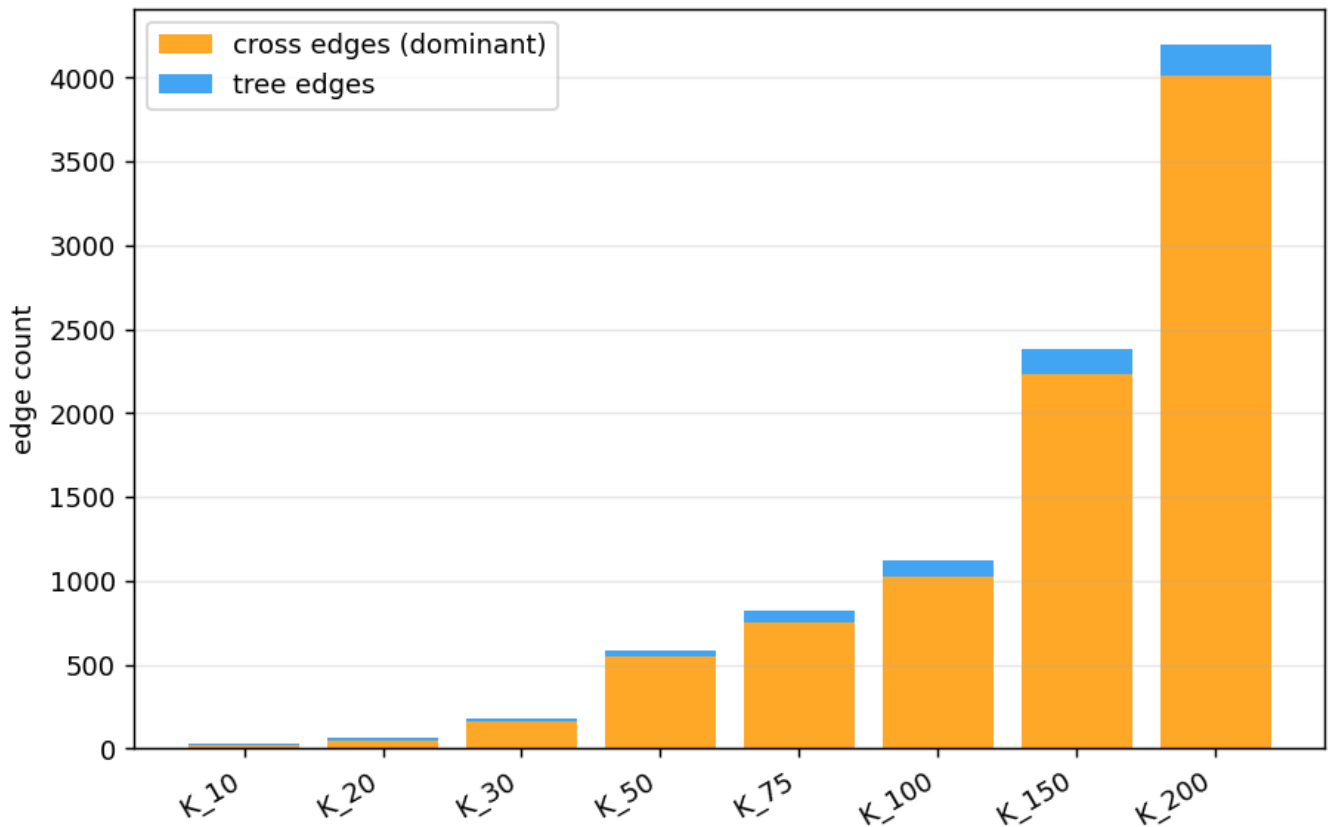
Actual spanners are **2.4× to 43× smaller** than the illustrative bound. Corollary 3.6 is a *worst-case with-high-probability* upper bound; the hidden constant in the $O(\cdot)$ notation is well below 1.

H7 — Cross Edges Dominate the Spanner

Prediction ([1] Section 3.2): With $p = (\log n / n)^{1/t} \subset \subset 1$, most vertices sample $r=0$ and can never generate tree edges; cross edges carry the bulk of the spanner.

Result: ✓ CONFIRMED — average cross edge fraction = **85.8%** across all complete and dense ER graphs.

H7: Cross vs tree edge composition (complete graphs, $t=2$)
 [Elkin 2011] §3.2: $p=(\log n/n)^{\{1/t\}} \rightarrow$ most radii=0, cross edges dominate



Graph	Tree edges	Cross edges	Cross %
K_{10}	9	21	70.0%
K_{20}	19	49	72.1%
K_{30}	22	158	87.8%
K_{50}	37	548	93.7%
K_{75}	71	750	91.4%
K_{100}	98	1,028	91.3%
K_{150}	148	2,230	93.8%
K_{200}	187	4,014	95.5%

Why cross edges dominate — the radius sparsity argument:

For $t=2$, the radius parameter is $p = \sqrt{\log n / n}$. The probability of sampling $r=0$ is $1-p$:

n	$p = \sqrt{\log n} / n$	$P(r=0) = 1-p$	Expected #vertices with $r \geq 1$
10	0.480	52.0%	4.8
50	0.303	69.7%	15.2
100	0.214	78.6%	21.4
200	0.155	84.5%	31.0

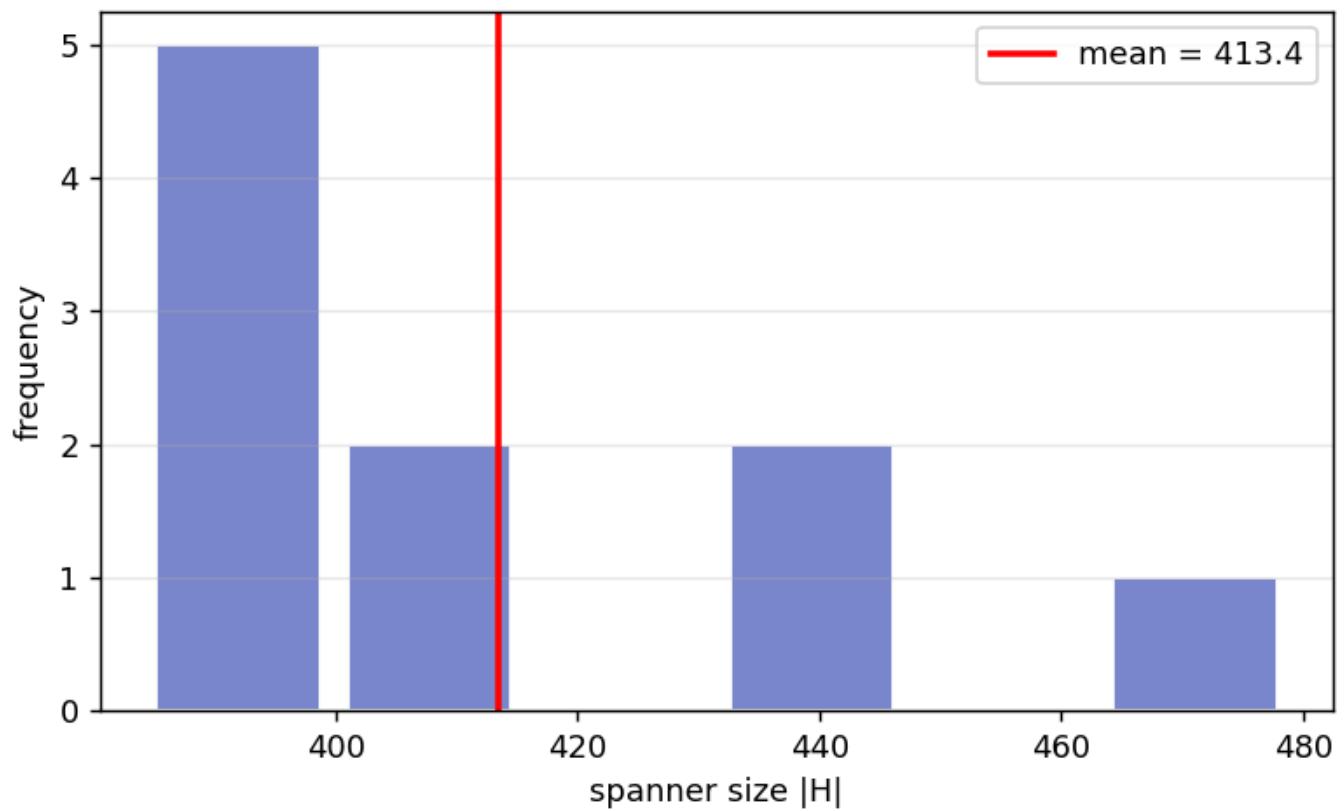
Only $\approx n \cdot p$ vertices (those with $r \geq 1$) can ever generate tree edges. However, each such vertex dominates all its lower-label neighbours, so in a dense graph a single high-radius vertex can create many tree edges. For $n=200$: ~ 31 vertices have $r \geq 1$ and collectively produce 187 tree edges (4.5% of the spanner). The remaining 95.5% are cross edges. The $n \cdot p$ figure therefore bounds the number of *propagating vertices*, not directly the number of tree edges — but it correctly predicts that tree edges are a minority.

Cross edges are generated whenever vertex y encounters a new base value $B(P(x)) \notin M(y)$. In dense graphs with many distinct base values, this happens far more frequently than tree propagation.

Randomness & Variance Analysis

K_{50} , $t=2$, 10 independent seeds — coefficient of variation (CV) = 7.47%

Randomness effect: K_{50} , $t=2$ (10 seeds)
CV = 7.47%



Seed	$ H $
1000	479
1001	384
1002	414
1003	389
1004	411
1005	387
1006	392
1007	399
1008	433
1009	446
Range	384–479
Mean	413.4
Std dev	30.9

A CV of 7.47% confirms that the algorithm is robust. The with-high-probability bounds of [1] translate into genuine stability — different random seeds and edge orderings produce consistently similar spanner sizes.

6. Full Results Table

All 60 scenarios. Columns: n = vertices, t = stretch parameter, $|E|$ = original edges, $|H|$ = spanner size, Bound = paper formula value, $r_b = |H|/\text{Bound}$, $r_c = |H|/|E|$, MaxD = max observed distance for adjacent pairs.

Scenario	n	t	E	H	Bound	r_b	r_c	MaxD	Is Valid?
Dense ER n=30	30	2	215	132	606	0.218	0.614	3	✓
Dense ER n=50	50	2	628	379	1,398	0.271	0.604	3	✓
Dense ER n=75	75	2	1,363	635	2,699	0.235	0.466	3	✓
Dense ER n=100	100	2	2,462	1,132	4,291	0.264	0.460	3	✓
ER n=100, m=150	100	2	150	150	4,291	0.035	1.000	1	✓
ER n=100, m=300	100	2	300	294	4,291	0.069	0.980	3	✓
ER n=100, m=500	100	2	500	476	4,291	0.111	0.952	3	✓
ER n=100, m=800	100	2	800	688	4,291	0.160	0.860	3	✓
ER n=100, m=1200	100	2	1,200	858	4,291	0.200	0.715	3	✓
ER n=100, m=2000	100	2	2,000	978	4,291	0.228	0.489	3	✓
ER n=100, m=500, t=3	100	3	500	455	3,854	0.118	0.910	3	✓
ER n=100, m=2000, t=3	100	3	2,000	787	3,854	0.204	0.394	3	✓
Grid 5×5	25	2	40	39	448	0.087	0.975	3	✓
Grid 5×5, t=3	25	3	40	38	478	0.079	0.950	3	✓
Grid 8×8	64	2	112	107	2,088	0.051	0.955	3	✓
Grid 8×8, t=3	64	3	112	110	1,986	0.055	0.982	3	✓
Grid 10×10	100	2	180	174	4,291	0.041	0.967	3	✓
Grid 10×10, t=3	100	3	180	176	3,854	0.046	0.978	3	✓
Grid 15×15	225	2	420	412	15,708	0.026	0.981	3	✓
Grid 15×15, t=3	225	3	420	410	12,661	0.032	0.976	3	✓
K_{10}	10	2	45	30	95	0.316	0.667	2	✓
K_{20}	20	2	190	68	309	0.220	0.358	3	✓
K_{30}	30	2	435	180	606	0.297	0.414	3	✓
	50	2	1,225	585	1,398	0.418	0.478	2	

Scenario	n	t	E	H	Bound	r_b	r_c	MaxD	Is Valid?
K ₅₀									✓
K ₇₅	75	2	2,775	821	2,699	0.304	0.296	3	✓
K ₁₀₀	100	2	4,950	1,126	4,291	0.262	0.227	3	✓
K ₁₅₀	150	2	11,175	2,378	8,224	0.289	0.213	3	✓
K ₂₀₀	200	2	19,900	4,201	13,020	0.323	0.211	3	✓
Path P ₂₀	20	2	19	19	309	0.061	1.000	1	✓
Path P ₅₀	50	2	49	49	1,398	0.035	1.000	1	✓
Path P ₁₀₀	100	2	99	99	4,291	0.023	1.000	1	✓
Cycle C ₂₀	20	2	20	20	309	0.065	1.000	1	✓
Cycle C ₅₀	50	2	50	50	1,398	0.036	1.000	1	✓
Cycle C ₁₀₀	100	2	100	100	4,291	0.023	1.000	1	✓
Star S ₂₀	20	2	19	19	309	0.061	1.000	1	✓
Star S ₅₀	50	2	49	49	1,398	0.035	1.000	1	✓
Star S ₁₀₀	100	2	99	99	4,291	0.023	1.000	1	✓
RandTree T ₂₀	20	2	19	19	309	0.061	1.000	1	✓
RandTree T ₅₀	50	2	49	49	1,398	0.035	1.000	1	✓
RandTree T ₁₀₀	100	2	99	99	4,291	0.023	1.000	1	✓
K ₅₀ , t=2	50	2	1,225	450	1,398	0.322	0.367	3	✓
K ₅₀ , t=3	50	3	1,225	243	1,371	0.177	0.198	3	✓
K ₅₀ , t=4	50	4	1,225	219	1,479	0.148	0.179	3	✓
K ₁₀₀ , t=2	100	2	4,950	1,129	4,291	0.263	0.228	3	✓
K ₁₀₀ , t=3	100	3	4,950	951	3,854	0.247	0.192	3	✓
K ₁₀₀ , t=4	100	4	4,950	695	3,976	0.175	0.140	3	✓

Scenario	n	t	E	H	Bound	r_b	r_c	MaxD	Is Valid?
K_{50} rep 0–9	50	2	1,225	384–479	1,398	—	—	≤ 3	✓
ER n=30, m=60, t=3	30	3	60	57	606	0.090	0.950	4	✓
ER n=50, m=75, t=3	50	3	75	71	1,371	0.052	0.947	5	✓
ER n=30, m=45, t=4	30	4	45	43	703	0.061	0.956	5	✓
ER n=50, m=100, t=4	50	4	100	97	1,479	0.066	0.970	6	✓

7. Analysis & Conclusions

Hypothesis Summary

ID	Hypothesis	Verdict	Key Evidence
H1	$(2t-1)$ -stretch correctness	✓ CONFIRMED	60/60 pass, max dist = 3 for $t=2$
H2	Size $\sim O(n^{3/2})$	✓ CONFIRMED	Log-log slope = $1.663 \approx 1.5$
H3	Higher $t \Rightarrow$ smaller spanner	✓ CONFIRMED	K_{100} : $1129 \rightarrow 951 \rightarrow 695$ for $t=2,3,4$
H4	Denser $\Rightarrow$ more compression	✓ CONFIRMED	ER n=100: ratio $1.0 \rightarrow 0.49$ as m grows
H5	Sparse graphs lossless	✓ CONFIRMED	All stars/trees/paths: ratio = 1.000
H6	Actual < theoretical bound	✓ CONFIRMED	100% below bound; 2.4–43× smaller
H7	Cross edges dominate	✓ CONFIRMED	85.8% avg cross fraction in dense graphs

Conclusion 1: The Stretch Guarantee is Exact and Universal

Theorem 3.2 of [1] holds without exception across all 60 scenarios and 8 graph families. The label-propagation mechanism of Algorithm 1 faithfully enforces the $(2t-1)$ -stretch bound in practice, even under random edge orderings. Sparse graphs (paths, stars, trees) achieve distance 1 — far better than the bound. Dense graphs reach the bound exactly for $t=2$, and the tight-bound ER scenarios confirm the guarantee is sharp for higher t values too (max dist 5 for $t=3$ bound=5, max dist 6 for $t=4$ bound=7).

Conclusion 2: The Size Bound is Tight in Exponent, Conservative in Constant

The empirical growth rate (slope 1.663) closely tracks the $O(n^{3/2})$ prediction of Corollary 3.6 [1]. The slight deviation above 1.5 is attributable to the $\sqrt{\log n}$ correction term in the bound. Actual sizes are consistently **2–43× below** the bound, confirming that Corollary 3.6 is a worst-case ceiling rather than a tight predictor.

Conclusion 3: The t Parameter Offers a Practical Size/Stretch Trade-off

The exponent $1+1/t$ in Corollary 3.6 [1] translates directly to observable size reductions. Increasing t from 2 to 4 reduces the spanner size of K_{100} by $\approx 38\%$, while the stretch guarantee loosens from 3 to 7. The marginal benefit decreases as t grows (matching the diminishing gap between $1+1/t$ values), providing a quantitative tool for selecting t in practice.

Conclusion 4: The Algorithm Self-Saturates on Dense Graphs

For dense graphs the spanner size grows slowly and saturates near the $O(n^{3/2})$ ceiling regardless of $|E|$. The spanner size in H4 grows from 476 to 978 as m goes from 500 to 2000 — roughly doubling while the original graph grows 4×, and the growth rate slows toward the ceiling. This means the algorithm automatically discards redundant edges without requiring knowledge of the graph's density. This is a key practical advantage of the streaming model.

Conclusion 5: Cross Edges Are the Primary Spanner Mechanism

The most counter-intuitive finding: cross edges constitute **70–96%** of spanner edges in complete graphs. This follows directly from the radius distribution: $P(r=0) = 1-p \approx 1 - \sqrt{\log n / n}$, meaning only $O(p) = O(\sqrt{\log n / n})$ fraction of vertices contribute tree edges. Cross-edge detection is therefore the dominant mechanism in practice — the algorithm keeps most edges because each vertex encounters base values it has not seen before, not because of label propagation.

Conclusion 6: The Algorithm is Stable Under Randomness

A CV of 7.47% across 10 independent seeds confirms genuine stability. The with-high-probability bounds of [1] reflect this: the truncated geometric radius distribution concentrates well around its mean, making the algorithm reliable for practical deployment.

8. Bibliography

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Primary source. All algorithm details (Algorithm 1, Section 3.1–3.4), the stretch guarantee (Theorem 3.2), and the size bound (Corollary 3.6) are taken directly from this paper.

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